

ANN8: Hopfield Network (2 Vectors)

Cell 1: Import Library

Code:

```
import numpy as np
```

Explanation (what to say in viva):

Import numpy for matrix operations.

Hopfield network relies on matrix multiplication.

Cell 2: Define Patterns

Code:

```
x = np.array([[1, -1, 1, -1], [-1, 1, -1, 1]])
```

Explanation (what to say in viva):

These are stored patterns.

Hopfield stores patterns as memory.

Using bipolar values improves performance.

Cell 3: Weight Matrix

Code:

```
w = np.zeros((4,4))
for p in X:
    w += np.outer(p,p)
np.fill_diagonal(w,0)
```

Explanation (what to say in viva):

Outer product stores pattern.

Diagonal set to zero removes self-connection.

This follows Hebbian learning rule.

Cell 4: Activation Function

Code:

```
def sign(x): return np.where(x>=0,1,-1)
```

Explanation (what to say in viva):

Converts values to +1 or -1.

Used to update neuron state.

Cell 5: Update Rule

Code:

```
x = sign(np.dot(w,x))
```

Explanation (what to say in viva):

Multiply weight matrix with input.

Apply sign function.

Repeat until stable state is reached.

Cell 6: Output

Code:

```
print(x)
```

Explanation (what to say in viva):

Displays recovered pattern.

Should match stored pattern.